

Jesse He

Curriculum Vitae

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Education

Halicioğlu Data Science Institute, UC San Diego, San Diego, California

PhD Student, Advised by Yusu Wang, Gal Mishne

The Ohio State University, Columbus, OH

Bachelor of Science in Mathematics (Honors), *magna cum laude* May 2022

Bachelor of Arts in Computer and Information Science, *magna cum laude* May 2022

Research Interests

My research interests lie in the intersection of interpretable machine learning and geometric data analysis. In particular, I am interested in explainability methods for graph neural networks and interpretable manifold learning.

Publications

D. Kohli, **J. He**, C. Holtz, G. Mishne, and A. Cloninger, *Robust estimation of boundary using doubly stochastic scaling of Gaussian kernel*, Preprint, 2024. arXiv: 2411.18942 [math.ST].

J. He, H. Jenne, H. Chau, D. Brown, M. Raugas, S. Billey, and H. Kvinge, *Machines and mathematical mutations: Using GNNs to characterize quiver mutation classes*, Preprint. Accepted to ICML 2025. arXiv: 2411.07467 [cs.LG].

H. Chau, H. Jenne, D. Brown, **J. He**, M. Raugas, S. Billey, and H. Kvinge, *Machine learning meets algebraic combinatorics: A suite of datasets capturing research-level conjecturing ability in pure mathematics*, Preprint. Accepted to ICML 2025 (Oral). arXiv: 2503.06366 [cs.LG].

J. He, T. Brugère, and G. Mishne, “Product manifold learning with independent coordinate selection,” in *Proceedings of 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML)*, 2023. Available: <https://proceedings.mlr.press/v221/he23a.html>.

Employment

Research

SU2024- **National Security Internship, AI and Data Analytics**, Pacific Northwest National Laboratory, Richland, WA (remote)
Supervised by Henry Kvinge

SU2021 **Emerging Issues in Cybersecurity REU**, New Mexico Tech, Socorro, NM (remote)
Supervised by Subhasish Mazumdar

Teaching

Graduate Teaching Assistant, UC San Diego, Halicioğlu Data Science Institute
WI 2024 DSC 206: Algorithms for Data Science

Undergraduate Grader, The Ohio State University Department of Mathematics

SP2022 Math 3345H: Honors Foundations of Higher Mathematics

SP2022 Math 5591H/5112: Honors Abstract Algebra II

AU2021 Math 5590H/5111: Honors Abstract Algebra I

Undergraduate Grader, *The Ohio State University Department of Computer Science and Engineering*

AU2020-SP2021 CSE 3521: Survey of Artificial Intelligence I

AU2019 CSE 2221: Software Components

Digital Sandbox Project Group Instructor, *The Ohio State University Media, Marketing, and Communications Scholars*

SP2020 Introduction to L^AT_EX

Awards

Qualcomm Innovation Fellowship 2024 Winner

Skills

Programming Python, Matlab, R, C, C++, C#, Java

Other Git, L^AT_EX, Max/MSP/Jitter, Cockos REAPER

Languages

Mandarin Conversational

Japanese Basic

Spanish Basic